LAWS OF MOTION

Newton’s Laws of Motion

Newton’s First Law:

A body remains at rest or in uniform motion unless acted upon by an external force.

Example:

A passenger falls backward when a bus suddenly starts due to inertia of rest.

Newton’s Second Law:

The rate of change of momentum is directly proportional to the applied force and takes place in the direction of force.

Formula:

$F = ma$

Where:

- F = Force
- m = Mass
- a = Acceleration

SI Unit of Force:

Newton (N)

Important Point:

Greater the mass of a body, greater is the force required to change its motion.

Momentum:

$$p = mv$$

Where:

- p = Momentum
- m = Mass

- v = Velocity

Newton’s Third Law:

For every action, there is an equal and opposite reaction.

Example:

While walking, we push the ground backward and the ground pushes us forward.

Conservation of Momentum:

When no external force acts on a system, the total momentum of the system remains constant.

Formula:

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Where:

- m_1, m_2 = Masses of bodies
- u_1, u_2 = Initial velocities
- v_1, v_2 = Final velocities

Impulse:

Impulse is the product of force and time interval for which it acts.

Formula:

Impulse = $F \times t$

Applications of Newton’s Laws

Seat Belts:

Seat belts protect passengers from sudden jerks due to inertia.

Rocket Motion:

Rocket propulsion works on Newton’s Third Law of Motion.

Pulling a Carpet:

When a carpet is pulled quickly, dust particles remain at rest due to inertia.

Quick Revision Points:

- Force changes the state of motion of a body.
- Inertia is the resistance to change in motion.
- Newton’s First Law explains inertia.
- Newton’s Second Law gives relation F = ma.
- Newton’s Third Law states action and reaction.
- Total momentum remains conserved if no external force acts.
- Impulse = Force × Time

Important Formulae:

$F = ma$

$p = mv$

Impulse = $F \times t$